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PROFILE

Software engineer with production experience and four shipped projects, currently transitioning into formal AI/ML training. Builds AI products end-to-end, from OpenAI and Whisper integrations to offline-first mobile architectures with native iOS engineering. First Class Honours graduate from the University of East London (85%), applying to Imperial MSc Computing (Artificial Intelligence and Machine Learning) for 2026 entry.

EDUCATION

University of East London Sept 2024 – Jun 2025

BSc (Hons) Applied Computing — First Class Honours, 85%

Relevant modules: Enterprise Architecture and Cloud Computing (CN6000), Computer and Network Security (CN6003), Project Management (CN6004).

FAST National University of Computer and Emerging Sciences (NUCES) Sept 2020 – Jun 2024

BSc Artificial Intelligence (transferred from BSc Computer Science in 2021)

Relevant modules: Artificial Intelligence, Machine Learning, Probability and Statistics, Data Structures.

A separate Extenuating Circumstances Statement provides context for my FAST academic record.

EXPERIENCE

Kaye MacKenzie — Manchester, UK Jun 2025 – Nov 2025

Software Engineer Intern

- Led a full backend migration from MongoDB to Supabase (PostgreSQL) on a cross-platform retail inventory and invoicing app, redesigning schemas and rewriting REST APIs end-to-end; reduced operational cost by ~30% with no impact on load times.
- Built frontend features and supporting backend APIs on **IntellectraX**, a Next.js EdTech web platform, shipping production-ready updates on a weekly release cadence.
- Worked across TypeScript, Next.js, React Native, Node.js, and Supabase in production, collaborating with product and design to ship reliability, performance, and UX improvements.

SELECTED PROJECTS

Khushoo — Cross-platform Prayer Tracking App with Live Social Widgets React Native, Expo SDK 54, TypeScript, Supabase, SQLite,

Swift/WidgetKit (private; launching 2026)

- Engineered a live-widget pipeline using a Postgres trigger on prayer_logs invoking a pg_net.http_post edge function, dispatching silent push notifications that wake peers' apps and refresh native iOS and Android widgets in near-real time. No major social application currently maintains this capability.
- Architected an offline-first sync system with SQLite as local source of truth and Supabase as cloud backend; implemented optimistic UI via React Query, a retry-aware sync queue (max 5 attempts per item), and debounced profile writes for resilience on unreliable networks.
- Developed 6 native iOS widgets in Swift/WidgetKit and 10 Android widgets via react-native-android-widget, sharing data across app and widget targets through an App Group store and a custom Expo native module.
- Diagnosed a release-build constraint in Apple's AppIntents framework, where reflection only inspects public types, per WWDC23, unblocking the entire interactive-widget feature after two days of tracing.
- Integrated Supabase Auth, Postgres with Row Level Security, and Realtime for social "Circles"; instrumented in production with PostHog and Sentry.

Fattys — Mobile Boutique Commerce App with Realtime Courier Tracking React Native, Expo SDK 54, TypeScript (strict), Supabase,

Zustand, React Query, EAS Build (private; in production)

- Engineered an atomic order pipeline via a create_order_transaction Postgres RPC that validates per-variant inventory, applies discount codes, inserts order + line items + status-history rows in a single transaction, and de-duplicates retries through a client-supplied idempotency_key.
- Designed a realtime courier-tracking timeline without polling: a PostEx webhook triggers a service-role insert into tracking_events, which fires a Supabase Realtime push (filtered by shipment_id) and merges into the active order screen on-device. Terminal events auto-flip orders.status.
- Solved cross-device cart sync between the RN app and a Next.js web companion via a shared cart_items table with hydrate-on-sign-in union-merge, debounced 400ms push from the Zustand store, and Realtime subscription for cross-device pull.
- Resolved a Zustand persist race where AsyncStorage rehydration silently overwrote server-merged state; gated all sync effects on persist.onFinishHydration to fix.
- Diagnosed a TestFlight production crash caused by EXPO_PUBLIC_* env vars not propagating to EAS cloud builds without explicit eas.json mirroring, and a Hermes bundle crash from babel-plugin-react-compiler + a lazy-font require(); locked the env-var contract for future builds.

Nuch Summarizer — AI-Powered Text & Audio Summaries Next.js 14, TypeScript, Supabase, OpenAI API, Whisper ·

github.com/Zalaann/Nuch-AI-Article-Summarizer

- Built an AI web app for text and audio summarisation with extractive and abstractive pipelines via the OpenAI ChatGPT API, configurable summary length, and bullet-point or comprehensive output formats.
- Integrated OpenAI Whisper for transcription, supporting MP3/WAV/M4A uploads and in-browser audio recording with automatic transcription-to-summary workflows.
- Implemented authentication, protected routes, and per-user history with Supabase, plus a responsive UI animated with Framer Motion.

Recello — Mobile Marketplace React Native Expo, TypeScript, Supabase, NativeWind

- Shipped a mobile marketplace for buying and selling smartphones with onboarding, authentication, search, saved items, and detailed listings using Expo Router.
- Implemented listing creation with multi-image upload, client-side image compression, and a responsive image gallery.
- Integrated Supabase for auth, persistent storage, and real-time messaging via Row Level Security policies and typed PostgREST APIs.

TECHNICAL SKILLS

AI / ML & Data: Python, NumPy, Pandas, scikit-learn, OpenAI ChatGPT API, OpenAI Whisper, model evaluation, feature engineering, recommendation systems, image-based inference pipelines

Languages: TypeScript, JavaScript, Python, SQL, Swift, C++

Frontend & Mobile: React, Next.js, React Native (Expo), Tailwind CSS, NativeWind, Redux, ShadCN UI, Framer Motion

Backend & Data: Node.js, Express.js, Supabase, REST APIs, WebSockets, PostgreSQL, MySQL, MongoDB, SQLite, Firebase, database indexing

Infrastructure & Tooling: Git, Docker, Linux, Azure, CI/CD, Nginx, Vercel, PostHog, Sentry, Postman

LANGUAGES

English (fluent · IELTS 7.0), Urdu (native).

References available on request.